EFY TESTED
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SMARTWATCH-LIKE HID
With Keyboard And Touchpad

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Previously, in March 2023 issue, we published a project for a smartwatch with an ESP32 board featuring Wi-Fi analysis and health monitoring capabilities. Building upon this, let us now create a smartwatch that can function as a wearable human interface device (HID). This watch can serve as both a keyboard and a mouse, providing control over phones, laptops, etc. It acts as a portable keyboard with mouse functionality, worn on the wrist.

This design utilises the Indusboard coin along with GC9A01 driver and CST816S capacitive touch driver-based display, perfectly suited for a round display. The display is connected to the Indusboard using SPI and I2C pins onboard. Refer to Fig. 1 for the author's prototype. The components required for this device are listed in Bill of Materials table.

Fig. 1: Author's prototype

Software coding

The code is created using Arduino and the TFTeSPI library, with the CST816S touch driver library driving the display and touchscreen. The HID library comes pre-installed with the ESP32S2 Board SDK, so only TFTeSPI and CST816 libraries need to be installed. Search for them in the Arduino library manager and install. Configure the display driver and SPI pins in the TFTeSPI library folder found in the Arduino library folder. Open the user config

BILL OF MATERIALS

Components	Description	Quantity
Indusboard	Development board	1
USB Type C	Adaptor	1
Round touch display	GC9A01 driver and CST816S capacitive touch driver	1

```
//#define ILI9481_DRIVER
//#define ILI9486_DRIVER
//#define ILI9488_DRIVER     // WARNING: Do not connect ILI9488 display SDO to MISO if other devices
//#define ST7789_DRIVER      // Full configuration option, define additional parameters below for th
//#define ST7789_2_DRIVER    // Minimal configuration option, define additional parameters below for
//#define R61581_DRIVER
//#define RM68140_DRIVER
//#define ST7796_DRIVER
//#define SSD1351_DRIVER
//#define SSD1963_480_DRIVER
//#define SSD1963_800_DRIVER
//#define SSD1963_800ALT_DRIVER
//#define ILI9225_DRIVER         //////////////////////////////////////////////////////////////////
#define GC9A01_DRIVER

// Some displays support SPI reads via the MISO pin, other displays have a single
// bi-directional SDA pin and the library will try to read this via the MOSI line.
// To use the SDA line for reading data from the TFT uncomment the following line:

 #define TFT_SDA_READ      // This option is for ESP32 ONLY, tested with ST7789 and GC9A01 display c

// For ST7735, ST7789 and ILI9341 ONLY, define the colour order IF the blue and red are swapped on y
// Try ONE option at a time to find the correct colour order for your display

//  #define TFT_RGB_ORDER TFT_RGB  // Colour order Red-Green-Blue
//  #define TFT_RGB_ORDER TFT_BGR  // Colour order Blue-Green-Red

// For M5Stack ESP32 module with integrated ILI9341 display ONLY, remove // in line below

// #define M5STACK

// For ST7789, ST7735, ILI9163 and GC9A01 ONLY, define the pixel width and height in portrait orient
// #define TFT_WIDTH  80
// #define TFT_WIDTH  128
// #define TFT_WIDTH  172 // ST7789 172 x 320
// #define TFT_WIDTH  170 // ST7789 170 x 320
// #define TFT_WIDTH  240 // ST7789 240 x 240 and 240 x 320
// #define TFT_HEIGHT 160
// #define TFT_HEIGHT 128
```

Fig. 2: Setting the display driver